

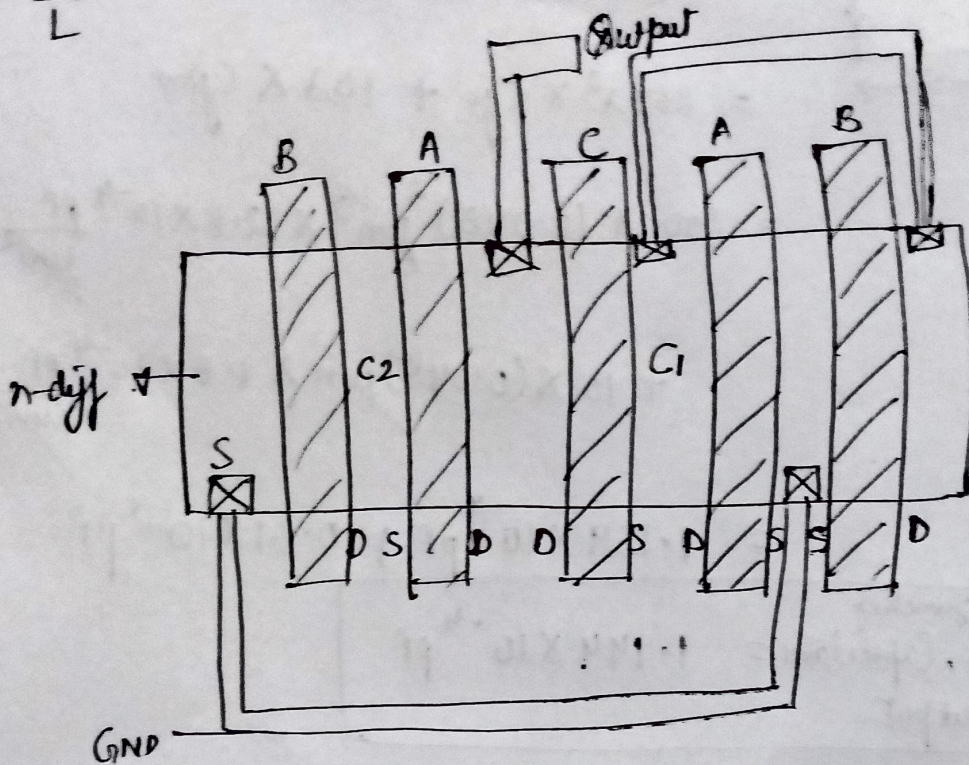
Solution-

1- $L = 2\lambda = 90 \text{ nm}$

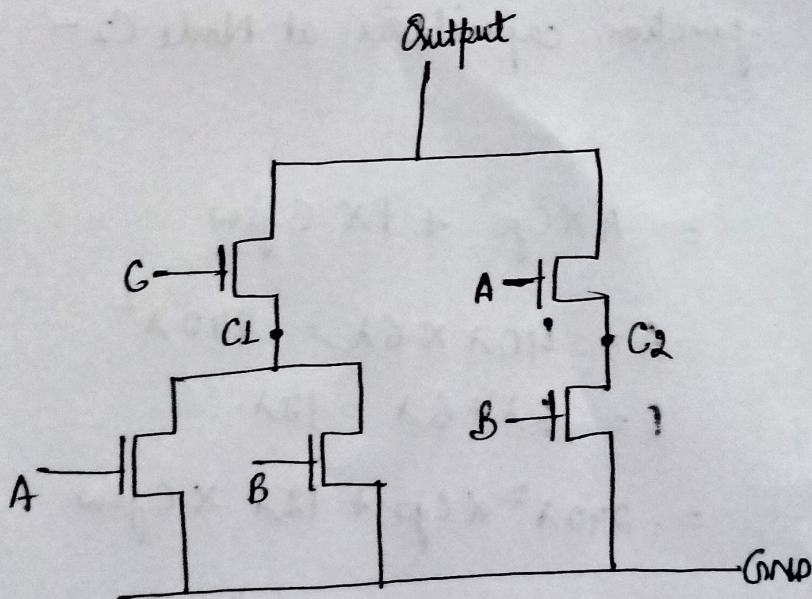
$\lambda = 45 \text{ nm}$

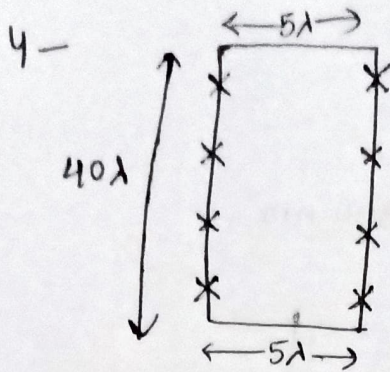
$\frac{W}{L} = 20 \Rightarrow W = 40\lambda = 40 \times 45 = 1800 \text{ nm}$

2-



3-





$$= A \times C_{jo} + P \times C_{jsw}$$

$$\text{Area} = A = 40\lambda \times 5\lambda = 200\lambda^2$$

$$\text{Perimeter} = P = 10\lambda$$

$$= 200\lambda^2 \times C_{jo} + 10\lambda \times C_{jsw}$$

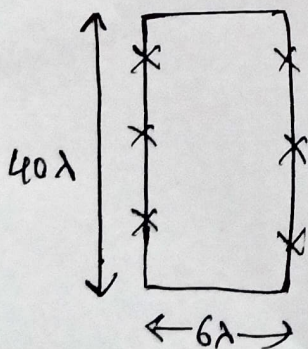
$$= 200 \times (0.045)^2 \mu\text{m}^2 \times 2.8 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}^2}$$

$$+ 10 \times (0.045) \mu\text{m} \times 1.8 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}}$$

$$= 1.134 \times 10^{-4} \text{ pF} + 0.81 \times 10^{-4} \text{ pF}$$

junction
Parasitic Capacitance = $1.944 \times 10^{-4} \text{ pF}$
at Output

5- Parasitic junction capacitance at Node C2-



$$= A \times C_{jo} + P \times C_{jsw}$$

$$A = 40\lambda \times 6\lambda = 240\lambda^2$$

$$P = 6\lambda + 6\lambda = 12\lambda$$

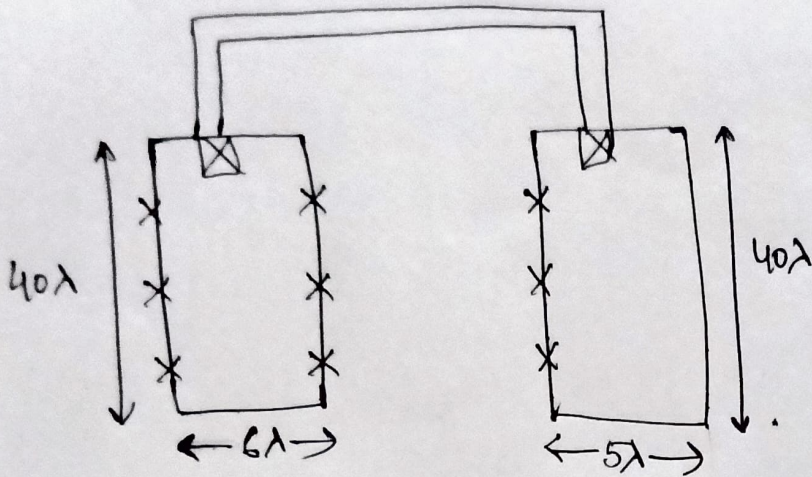
$$= 240\lambda^2 \times C_{jo} + 12\lambda \times C_{jsw}$$

$$= 240 \times (0.045)^2 \mu\text{m}^2 \times 2.8 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}^2} +$$

$$12 \times (0.045) \mu\text{m} \times 1.8 \times 10^{-4} \frac{\text{pF}}{\mu\text{m}}$$

Parasitic junction capacitance at Node C2 = $2.3328 \times 10^{-4} \text{ pF}$

Parasitic junction capacitance at Node C1 -



$$= A \times C_{jo} + P \times C_{jsw}$$

$$A = 240 \lambda^2 + 200 \lambda^2 = 440 \lambda^2$$

$$P = 5\lambda + 5\lambda + 6\lambda + 6\lambda + 40\lambda = 62\lambda$$

$$= 440 \lambda^2 \times \frac{2.8 \times 10^{-4} \text{ pF}}{\mu\text{m}^2} + 62\lambda \mu\text{m} \times \frac{1.8 \times 10^{-4} \text{ pF}}{\mu\text{m}}$$

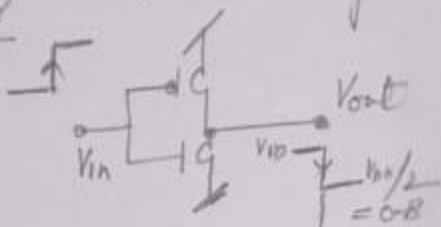
$$= 440 \times (0.045)^2 \times 2.8 \times 10^{-4} \text{ pF} + 62 \times (0.045) \times 1.8 \times 10^{-4} \text{ pF}$$

$$= 2.4948 \times 10^{-4} \text{ pF} + 5.022 \times 10^{-4} \text{ pF}$$

$$\boxed{\text{Parasitic junction capacitance at Node C1} = 7.5168 \times 10^{-4} \text{ pF}}$$

$$\textcircled{1} \tau_{HL} \approx 22 \text{ ps} = \frac{C_L \Delta V}{I_{avg}} = \frac{C_L (V_{DD}/2)}{I_{avg}} = \frac{24 \times 10^{-15}}{0.8} \times 0.8$$

$$\tau_{HL} \Rightarrow 22 \text{ ps} = \frac{24 \times 10^{-15} \times 0.8}{I_{avg}}$$



$$I_{avg} = \frac{1}{2} [I_{s,t} + I_{LH}]_{NMO5}$$

② s-t (NMO5)

$$I_{s,t} = \frac{1}{2} \times k_n (V_{gs} - V_t)^2 = \frac{1}{2} \times 280 \times 10^{-6} \times \frac{10}{L} (1.6 - 0.4)^2$$

$$I_{s,t} = 140 \times 10^{-6} \times \left(\frac{10}{L}\right) (1.2)^2$$

$$I_{s,t} = 201.6 \times \left(\frac{10}{L}\right) \times 10^{-6}$$

LINEAR ($V_{out} = V_{DD}/2$)

$$\textcircled{3} I_{D_{linear}} = \frac{1}{2} k_n [2(V_{gs} - V_t)V_{ds} - V_{ds}^2]$$

$$= \frac{1}{2} \times 280 \times 10^{-6} \times \left(\frac{10}{L}\right) [2(1.6 - 0.4) \times 0.8 - 0.8^2]$$

$$= 140 \times 10^{-6} \times \left(\frac{10}{L}\right) [1.92 - 0.64]$$

$$= 1.28$$

$$I_{D_{linear}} = 140 \times 10^{-6} \times 1.28 \times \left(\frac{10}{L}\right)$$

$$= 179.2 \times 10^{-6} \times \left(\frac{10}{L}\right)$$

0.036

$$I_{avg} = \frac{(201.6 + 179.2) \times 10^{-6}}{2} = 190.4 \times 10^{-6} \left(\frac{W}{L} \right)$$

$$\tilde{L} = \frac{C_L \Delta V}{I_{avg}} \Rightarrow 22 \times 10^{-12} = \frac{24 \times 10^{-15} \times 0.8}{190.4 \times 10^{-6} \times \left(\frac{W}{L} \right)}$$

$$\left(\frac{W}{L} \right) = \frac{24 \times 10^{-15} \times 0.8}{22 \times 10^{-12} \times 190.4 \times 10^{-6}} = \frac{24 \times 0.8 \times 10^{-3}}{22 \times 190.4 \times 10^{-6}}$$

$$\left(\frac{W}{L} \right) = \frac{4.5836 \times 10^{-3}}{10^{-6}} = \underline{4.58 \times 10^3}$$

$$\left(\frac{W}{L} \right)_H = 4.58 \times 10^3$$

Given: $H_n = 34p$

$$\left(\frac{W}{L} \right)_P = 3 \times 4.58 \times 10^3$$

$$\left(\frac{W}{L} \right)_P = 13.74 \times 10^3$$

$$b) V_{OH} = V_{DD}, V_{OL} = 0 \text{ (CMOS inverter)}$$

$$NM_L = V_{IL} - V_{OL} \quad \left| \quad NM_H = V_{OH} - V_{IH}\right.$$

$$V_{IL} = \frac{2V_{out} + V_{TOP} - V_{DD} + k_R V_{Ton}}{1 + k_R}$$

$$- k_R = \frac{k_n}{k_p} = \frac{\mu_n \epsilon_{ox} \left(\frac{W}{L}\right)}{\mu_p \epsilon_{ox} \left(\frac{W}{L}\right)} = \frac{3 \mu_p \epsilon_{ox} \left(\frac{W}{L}\right)}{\mu_p \epsilon_{ox} \cdot 3 \left(\frac{W}{L}\right)} \approx 1$$

$$V_{IL} = \frac{2 \times 1.5 + 0.4 - 1.6 + 0.4}{2} = \frac{2.2}{2} = \underline{1.1}$$

$$V_{IH} = \frac{V_{DD} + V_{TOP} + k_R (2V_{out} + V_{Ton})}{1 + k_R}$$

$$= \frac{1.6 + 0.4 + 1 (2 \times 1.5 + 0.4)}{2} = \frac{2 + 0.8}{2}$$

$$V_{IH} = \underline{1.4}$$

$$NM_L = V_{IL} - V_{OL} = 1.1 - 0 = \underline{1.1 V}$$

$$\bullet NM_H = V_{OH} - V_{IH} = 1.6 - 1.4 = \underline{0.2 V}$$

Voies
Niveau

$$E_{\text{supply}} = 24 \times 10^{-15} \times (1.6)^2$$

$$=$$

$$P_{\text{dyn}} = G V^2 f = 24 \times 10^{-15} \times (1.6)^2 \times \left(\frac{1}{22 \times 10^{12}} \right)$$

$$=$$

$$\text{G} \rangle E_{\text{supply}} = 24 \times 10^{-15} \times (1.6)^2$$

$$= 61.44 \times 10^{-15} \text{ Joule.}$$

$$P_{\text{dyn}} = G V^2 f = 24 \times 10^{-15} \times (1.6)^2 \times \left(\frac{1}{22 \times 10^{12}} \right)$$

$$P_{\text{dyn.}} = 61.44 \times 10^{-15} \times \frac{1}{22 \times 10^{12}}$$

$$P_{\text{dynamic}} = \underline{2.792 \times 10^{-3} \text{ W.}}$$

(3) (A)

$$R_p = \frac{1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_p (V_{DD} - |V_{tp}|)} = \frac{1}{90 \mu A/V^2 \times 8 \times (3 - 1.08)}$$

$$R_p = 631.3 \Omega$$

$$R_n = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L}\right)_n (V_{DD} - V_{tn})} = \frac{1}{140 \mu A/V^2 \times 4 \times (3 - 0.5)}$$

$$R_n = 714 \Omega$$

(B)

Worst-Case rise time for 2 i/p NAND gate:

$$t_r = 2.2 R_p C_{out} = 2.2 (631.3) \times 8 fF$$

$$t_r = 11.10 ps$$

Best-Case rise time for 2 i/p NAND gate:

$$t_r = 2.2 \left(\frac{R_p}{2}\right) \cdot C_{out} = 2.2 \left(\frac{631.3}{2}\right) \cdot 8 fF$$

$$t_r = 5.5 ps$$

Worst-Case fall time for 2 i/p NAND gate:

$$t_f = 2.2 (2 R_n C_{out} + R_n C_x)$$

$$t_f = 28.27 ps$$

Best-Case fall time for 2 i/p NAND gate:

$$t_f = 28.27 ps$$

(C)

NOR gate: - $P_1 = 0.25$, $P_0 = 0.75$
 switching activity factor $\alpha = P_0 P_1$
 $\alpha = 0.1875$

$$P_{dyn} = \alpha C_{out} V_{DD}^2 f = 13.5 \mu W$$

(d)

$$P_{dynamic} = \alpha_{av} \phi C_{av} V_{DD}^2$$

$$P_{Total} = P_{static} + P_{leakage} + P_{short-circuit} + P_{dynamic}$$

$$130 \text{ mW} = P_{\text{leakage}} + \underbrace{\alpha_{\text{av}} C_{\text{av}} V_{\text{DD}}^2 \phi}_K$$

$$130 \text{ mW} = P_{\text{leakage}} + K(15 \text{ MHz}) \quad \text{--- (1)}$$

$$100 \text{ mW} = P_{\text{leakage}} + K(10 \text{ MHz}) \quad \text{--- (2)}$$

$$\text{--- (1)} - \text{--- (2)}$$

$$K = \frac{30 \times 10^{-3}}{5 \times 10^6} = 6 \times 10^{-9}$$

From eq (1)

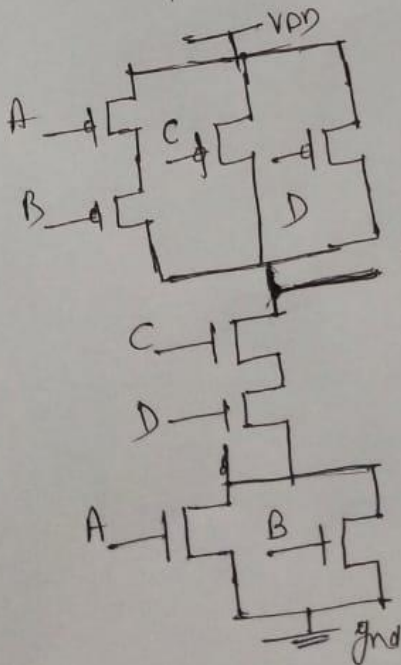
$$P_{\text{leakage}} = (130 - 90) \times 10^{-3} = 40 \text{ mW}$$

Q3B

$$Y = A'B + C' + D'$$

$$Y = \overline{A'B + C' + D'} = \overline{A'B} \cdot \overline{C'} \cdot \overline{D'}$$

$$Y = (A+B) \cdot C \cdot D$$



$$Y = A'B + C' + D'$$

Q4.

(a). Punch through will occur for a drain side depletion width = channel length – source side depletion width.

For the short channel case maximum drain side depletion width = 1 μm – 0.35 μm = 0.65 μm .

For the long channel case this should not ideally happen, however, it will occur for a drain side depletion width = channel length – source side depletion width.

Therefore, for the long channel case maximum drain side depletion width = 100 μm – 0.35 μm = 99.65 μm .

b).

Step 1 of 2

It is given that,

The MOS devices with short channel lengths do not follow square law so for them the current is given by,

$$I_D = WC_{ox}(V_{GS} - V_{TH})v_{sat} \dots\dots (1)$$

Where,

Gate capacitance per unit area is C_{ox}

Width of the channel is W

Potential difference between gate and silicon bar is $V_{GS} - V_{TH}$ Relative constant velocity is v_{sat}

Step 2 of 2

Calculate the transconductance of the MOS device.

$$\begin{aligned} g_m &= \frac{\delta I_D}{\delta V_{GS}} \\ &= \frac{\delta}{\delta V_{GS}} [WC_{ox}(V_{GS} - V_{TH})v_{sat}] \\ &= WC_{ox}v_{sat} \end{aligned}$$

Hence, the Transconductance of the given MOS device is $g_m = WC_{ox}v_{sat}$.

Q4:

c). Dynamic Power Dissipation, $P_D = \alpha C V_{DD}^2 f$

(After 30% Scaling Down)

$$PD' = \alpha C' V'^2 f'$$

$$PD' = P_D \times (0.7) \times 0.7^2 \times (1/0.7)$$

$$PD' = P_D \times 0.49$$

Therefore, the dynamic power dissipation is reduced to ~49% or reduced by 51%.

d). Since, $C = C_{ox} (WL)$ in $P_D = \alpha C V_{DD}^2 f$, expression

If gate oxide thickness is not scaled, C_{ox} is also not scaled.

$$\text{Therefore, } PD' = \alpha C' V'^2 f'$$

$$PD' = P_D \times (0.7)^2 \times 0.7^2 \times (1/0.7)$$

$$PD' = P_D \times 0.49 \times 0.7$$

$$PD' = P_D \times 0.343$$

Therefore, the dynamic power dissipation is reduced to ~34% or reduced by ~66%.